

Taylor-Proximal Approximation for Homoskedastic Matrix Noise

1 Model

We narrow the Taylor-proximal approximation to an isotropic Gaussian likelihood with unknown variance. Let $\Omega \subseteq \{1, \dots, m\} \times \{1, \dots, n\}$ be the observed entries, and let $g = \gamma^2$. The model is

$$Y_{ij} \mid X_{ij}, g \sim \mathcal{N}(X_{ij}, g), \quad (i, j) \in \Omega, \quad (1)$$

$$X \mid g, \lambda \sim \text{NND}\left(\frac{\lambda}{\sqrt{g}}\right), \quad (2)$$

$$g \sim P_g. \quad (3)$$

Equivalently, if $E = Y - X$, then $E_{ij} \mid g \sim \mathcal{N}(0, g)$ on observed entries. The scale factor $1/\sqrt{g}$ in the nuclear-norm prior is the homoskedastic analogue of the scale coupling used by the Lasso: larger noise weakens the effective penalty, while smaller noise strengthens it. Since the nuclear-norm distribution has normalizing constant proportional to η^{-mn} for parameter η , the prior density contributes

$$\log p(X \mid g, \lambda) = \text{const} + mn \log\left(\frac{\lambda}{\sqrt{g}}\right) - \frac{\lambda}{\sqrt{g}} \|X\|_*. \quad (4)$$

Thus the negative log density contains both $(\lambda/\sqrt{g})\|X\|_*$ and $(mn/2) \log g$. This scaling is the piece that keeps the posterior mode from degenerating toward the smallest admissible variance.

For the implemented version we use an inverse-gamma variance prior,

$$p(g) \propto g^{-(a_0+1)} \exp(-b_0/g), \quad (5)$$

although the objective below separates the prior contribution so that another one-dimensional prior on g can be substituted.

2 Taylor Delta Objective

Let μ denote the variational posterior mean for X , and define

$$B(\mu) = \left(\mu^\top \mu + \epsilon I\right)^{-1/2}, \quad D_i = \text{diag}(\mathbf{1}\{(i, j) \in \Omega\})_{j=1}^n. \quad (6)$$

At fixed (μ, g) the row precision induced by the Taylor-delta approximation is

$$A_i(\mu, g) = \frac{1}{g} D_i + \frac{\lambda}{\sqrt{g}} B(\mu). \quad (7)$$

Dropping constants independent of (μ, g) , the collapsed minimization objective is

$$\begin{aligned} \mathcal{J}(\mu, g) = & \frac{1}{2g} \sum_{(i,j) \in \Omega} (y_{ij} - \mu_{ij})^2 + \frac{1}{2} \sum_{i=1}^m \log |A_i(\mu, g)| \\ & + \frac{\lambda}{\sqrt{g}} \|\mu\|_* + \left(\frac{|\Omega|}{2} + \frac{mn}{2} + a_0 + 1 \right) \log g + \frac{b_0}{g}. \end{aligned} \quad (8)$$

The first $\log g$ term comes from the Gaussian likelihood, the second from the normalizing constant of the scaled nuclear-norm prior, and the final two terms from the inverse-gamma prior on g .

3 Optimization

The implementation optimizes μ and g by alternating monotone steps. For a fixed variance g , split (8) into a smooth part $h_g(\mu)$ and the nonsmooth nuclear-norm term $(\lambda/\sqrt{g})\|\mu\|_*$. A proximal-gradient step is

$$Z^{(t)} = \mu^{(t)} - s_t \nabla h_g(\mu^{(t)}), \quad (9)$$

$$\mu^{(t+1)} = \text{SVT}_{s_t \lambda / \sqrt{g}}(Z^{(t)}), \quad (10)$$

where SVT_τ soft-thresholds the singular values by τ . The step size s_t is selected by backtracking until $\mathcal{J}(\mu^{(t+1)}, g) \leq \mathcal{J}(\mu^{(t)}, g)$.

After the mean update, the scalar variance parameter is updated on the unconstrained scale $\ell = \log g$:

$$\ell^{(t+1)} = \ell^{(t)} - r_t \frac{\partial}{\partial \ell} \mathcal{J}(\mu^{(t+1)}, e^{\ell^{(t)}}), \quad (11)$$

again with backtracking to enforce monotonic decrease of $\mathcal{J}(\mu, e^\ell)$. Optimizing $\log g$ makes positivity automatic and keeps the variance update as a one-dimensional correction to the matrix proximal step.

4 Recovered Posterior Covariance

At the final iterate, the row covariance approximation is recovered using the same Taylor precision:

$$\Sigma_i = \left[\frac{1}{\hat{g}} D_i + \frac{\lambda}{\sqrt{\hat{g}}} B(\hat{\mu}) \right]^{-1}. \quad (12)$$

The returned effective penalty is

$$\lambda_{\text{eff}} = \frac{\lambda}{\sqrt{\hat{g}}}, \quad (13)$$

so downstream scoring code can reuse the same covariance and prior-variance interfaces as the known-variance Taylor approximation.